

Discrete Math Study Guide: Counting & Combinatorics

1. The Basic Counting Rules

- **Product Rule:** If a task has two independent steps, with m ways to do the first and n ways to do the second, there are $m \cdot n$ ways in total.
 - *Example:* A password is 1 letter followed by 1 digit. There are $26 \cdot 10 = 260$ possible passwords.
- **Sum Rule:** If a task can be done in one of m ways **or** one of n ways, with no overlap, there are $m + n$ ways.
 - *Example:* You choose 1 elective from 5 CS courses or 4 math courses:
 $5 + 4 = 9$ choices.

2. Permutations (Order Matters)

- **Formula:** The number of ways to arrange r items chosen from n is

- $$P(n, r) = \frac{n!}{(n-r)!}$$

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- **Example:** How many ways can 1st, 2nd, and 3rd place be awarded among 5 runners?

- $$P(5, 3) = \frac{5!}{2!} = \frac{120}{2} = 60$$

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- **Key idea:** Use permutations when swapping positions creates a *different* outcome, such as rankings, seating arrangements, or passwords.

3. Combinations (Order Doesn't Matter)

- **Formula:** The number of ways to choose r items from n is

- $$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

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- **Example:** How many ways can you pick a 3-person committee from 10 people?

- $$\binom{10}{3} = \frac{10!}{3!7!} = \frac{720}{6} = 120$$

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- **Relationship to permutations:** $P(n, r) = C(n, r) \cdot r!$, because each group of r items can be ordered in $r!$ ways.

4. Counting With Repetition

- **Ordered, repetition allowed:** n^r
 - *Example:* A 4-digit PIN has $10^4 = 10,000$ possibilities.
- **Unordered, repetition allowed ("stars and bars"):**

- $\binom{n+r-1}{r}$

- *Example:* You order 5 scoops of ice cream from 3 flavors, and repeats are allowed.

- $\binom{3+5-1}{5} = \binom{7}{5} = 21$

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5. Inclusion–Exclusion Principle

- **Two sets:**

- $|A \cup B| = |A| + |B| - |A \cap B|$

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- **Three sets:**

- $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

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- **Example:** How many integers from 1 to 100 are divisible by 3 or 5?
 - Divisible by 3: $\lfloor 100/3 \rfloor = 33$
 - Divisible by 5: $\lfloor 100/5 \rfloor = 20$
 - Divisible by both (that is, by 15): $\lfloor 100/15 \rfloor = 6$
 - Answer: $33 + 20 - 6 = 47$

6. Pigeonhole Principle

- **Basic form:** If $n + 1$ objects go into n boxes, at least one box holds 2 or more objects.
 - *Example:* In any group of 13 people, at least 2 share a birth month.
- **Generalized form:** If N objects go into k boxes, some box holds at least $\left\lceil \frac{N}{k} \right\rceil$ objects.

- *Example:* With 50 students and 7 days of the week, some day has at least $\lceil 50/7 \rceil = 8$ birthdays.

7. Binomial Theorem

- **Formula:**

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$$

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- **Example:** Find the coefficient of x^3 in $(x + 2)^5$.

- The general term is $\binom{5}{k} x^{5-k} \cdot 2^k$.
- For x^3 , you need $5 - k = 3$, so $k = 2$.
- The coefficient is $\binom{5}{2} \cdot 2^2 = 10 \cdot 4 = 40$.

- **Useful identity:** $\sum_{k=0}^n \binom{n}{k} = 2^n$, which is the number of subsets of an n -element set.

Practice Problems

1. How many ways can 8 books be arranged on a shelf? **Answer:** $8! = 40,320$
2. How many 5-card hands can be dealt from a 52-card deck? **Answer:** $\binom{52}{5} = 2,598,960$
3. How many integers from 1 to 60 are divisible by 2 or 3? **Answer:** $30 + 20 - 10 = 40$
4. What is the coefficient of $x^2 y^4$ in $(x + y)^6$? **Answer:** $\binom{6}{4} = 15$

Quick Decision Checklist

- **Does order matter?** If yes, use permutations. If no, use combinations.
- **Is repetition allowed?** If yes, use n^r when order matters, or stars and bars when it doesn't.
- **Are you counting overlapping groups?** Use inclusion–exclusion.
- **Are you proving something must happen?** Try the pigeonhole principle.

What topic do you need for your class? I can make the same kind of guide for logic, proofs and induction, sets and functions, relations, graph theory, recurrence relations, or number theory.